LAB 7 – FUZZ TESTING FOR INPUT VALIDATION BUGS

Food Delivery application | Evidence screenshots

Evidence screenshots

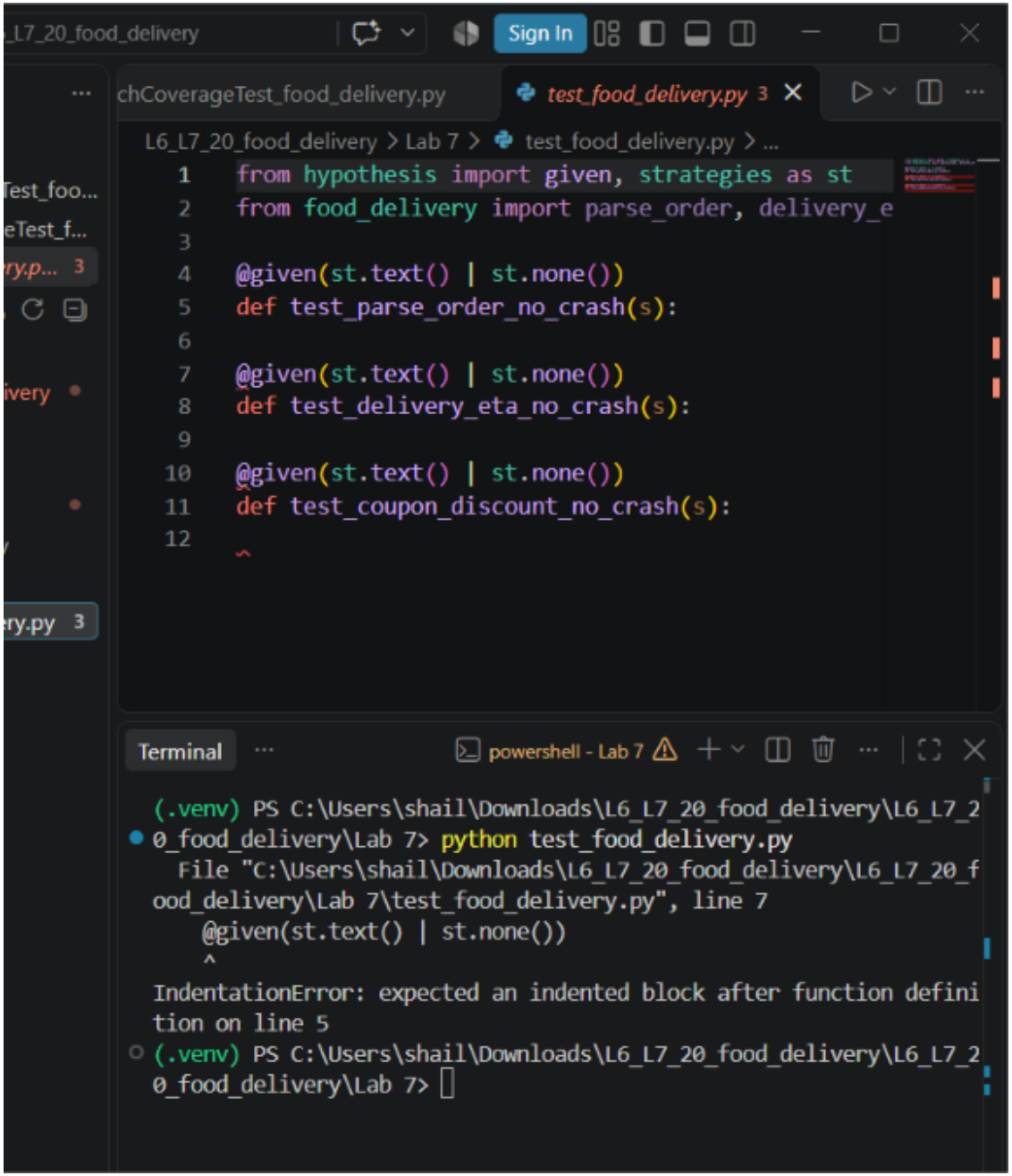
1. Original Lab 7 application execution

```
(.venv) PS C:\Users\shail\Downloads\L6_L7_20_food_delivery\L6_L7_20_food_delivery\Lab 7> python food_delivery.py
==== Running parse_order ====
Enter items as name:qty CSV (e.g. pizza:2,burger:1): pizza:3,burger:2
Order: {'pizza': 3, 'burger': 2}

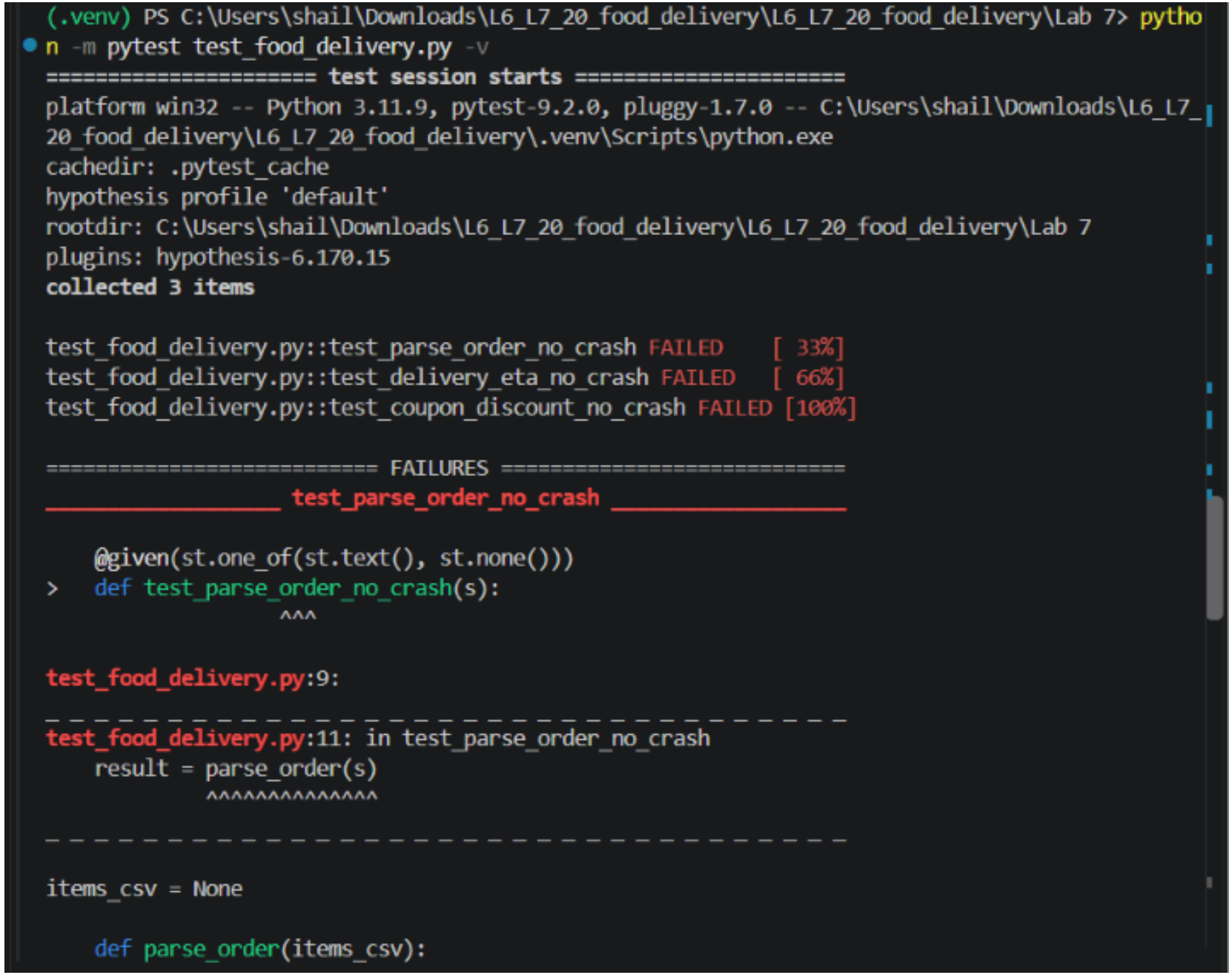
==== Running delivery_eta ====
Enter distance in km (e.g. 4.5): 4.5
ETA in Minutes: 33

==== Running coupon_discount ====
Enter a coupon code (e.g. SAVE50): SAVE 60
Discount: 60
(.venv) PS C:\Users\shail\Downloads\L6_L7_20_food_delivery\L6_L7_20_food_delivery\Lab 7>
```

2. Original Hypothesis tests – failures found in all three functions



3. Fuzz test failure detail – parse_order and delivery_eta



```
(.venv) PS C:\Users\shail\Downloads\L6_L7_20_food_delivery\L6_L7_20_food_delivery\Lab 7> pytho
n -m pytest test_food_delivery.py -v
==================== test session starts ====================
platform win32 -- Python 3.11.9, pytest-9.2.0, pluggy-1.7.0 -- C:\Users\shail\Downloads\L6_L7_
20_food_delivery\L6_L7_20_food_delivery\.venv\Scripts\python.exe
cachedir: .pytest_cache
hypothesis profile 'default'
rootdir: C:\Users\shail\Downloads\L6_L7_20_food_delivery\L6_L7_20_food_delivery\Lab 7
plugins: hypothesis-6.170.15
collected 3 items

test_food_delivery.py::test_parse_order_no_crash FAILED    [ 33%]
test_food_delivery.py::test_delivery_eta_no_crash FAILED   [ 66%]
test_food_delivery.py::test_coupon_discount_no_crash FAILED [100%]

========================== FAILURES ==========================
_____________________ test_parse_order_no_crash _____________________

    @given(st.one_of(st.text(), st.none()))
>   def test_parse_order_no_crash(s):
                ^^^

test_food_delivery.py:9:

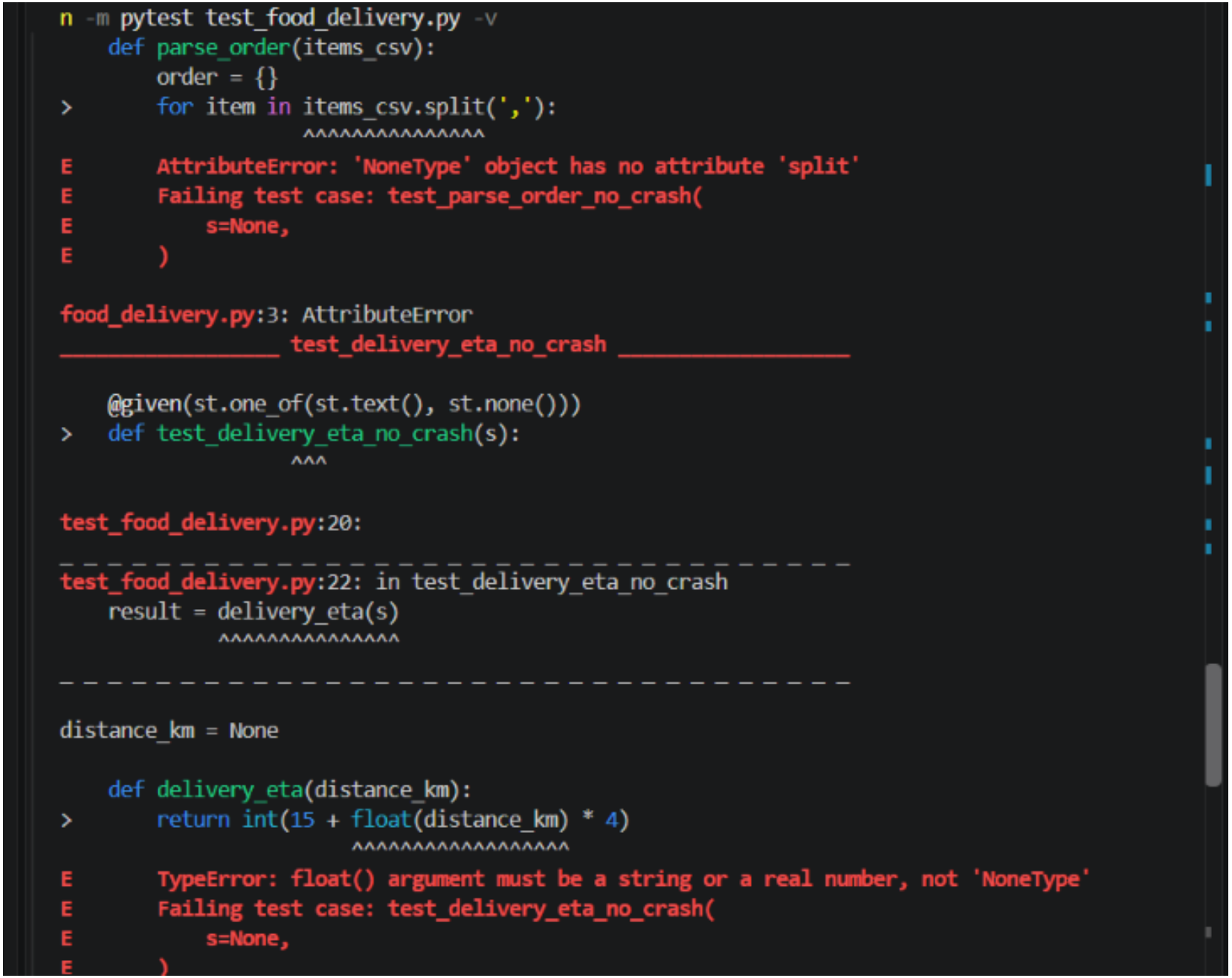
- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -
test_food_delivery.py:11: in test_parse_order_no_crash
    result = parse_order(s)
             ^^^^^^^^^^^^^^

- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -

items_csv = None

    def parse_order(items_csv):
```

4. Fuzz test failure detail – coupon_discount



```
n -m pytest test_food_delivery.py -v
    def parse_order(items_csv):
        order = {}
>       for item in items_csv.split(','):
                    ^^^^^^^^^^^^^^^^
E       AttributeError: 'NoneType' object has no attribute 'split'
E       Failing test case: test_parse_order_no_crash(
E           s=None,
E       )

food_delivery.py:3: AttributeError
_____________ test_delivery_eta_no_crash _____________

    @given(st.one_of(st.text(), st.none()))
>   def test_delivery_eta_no_crash(s):
                   ^^^

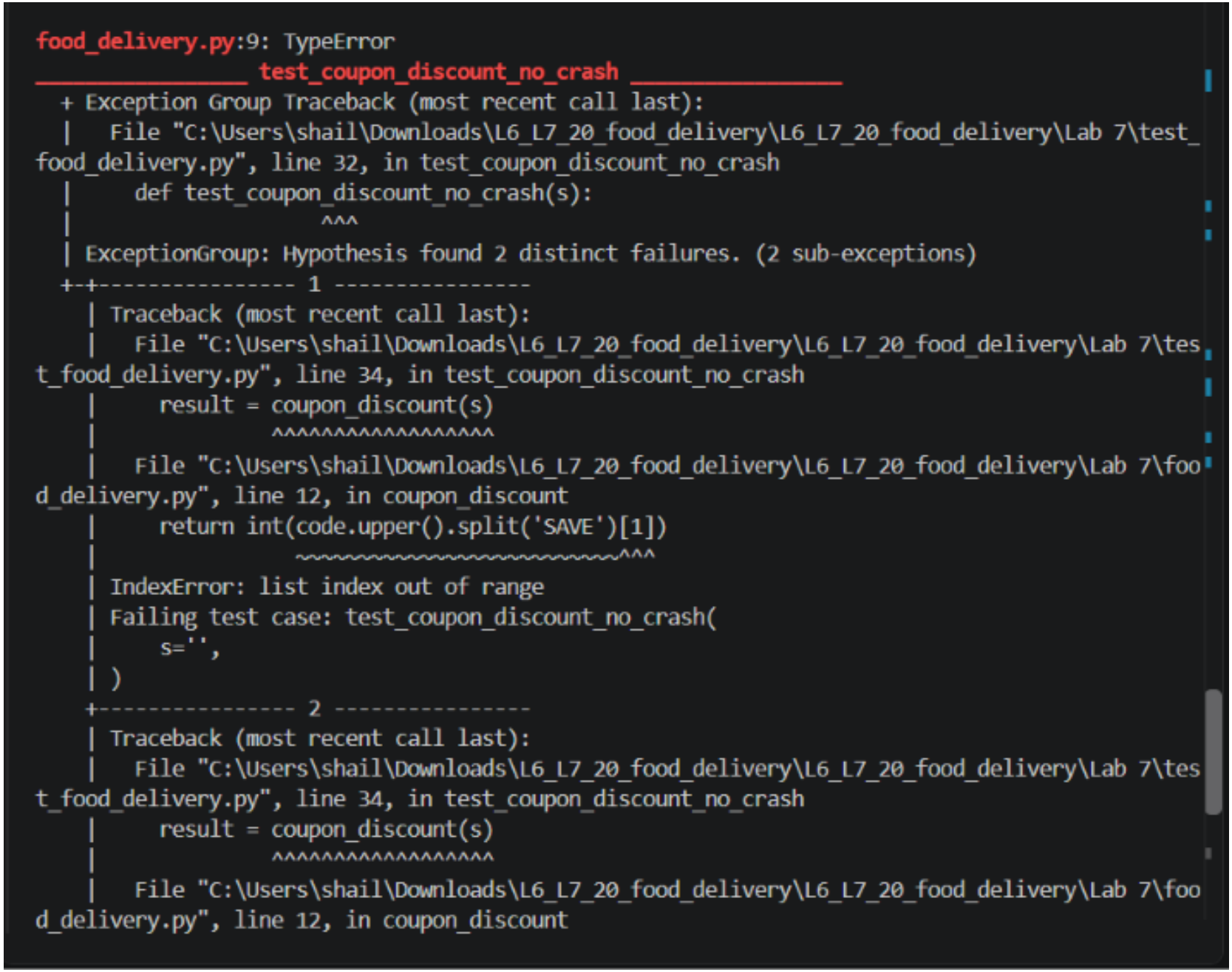
test_food_delivery.py:20:
_ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _
test_food_delivery.py:22: in test_delivery_eta_no_crash
    result = delivery_eta(s)
             ^^^^^^^^^^^^^^^

_ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _

distance_km = None

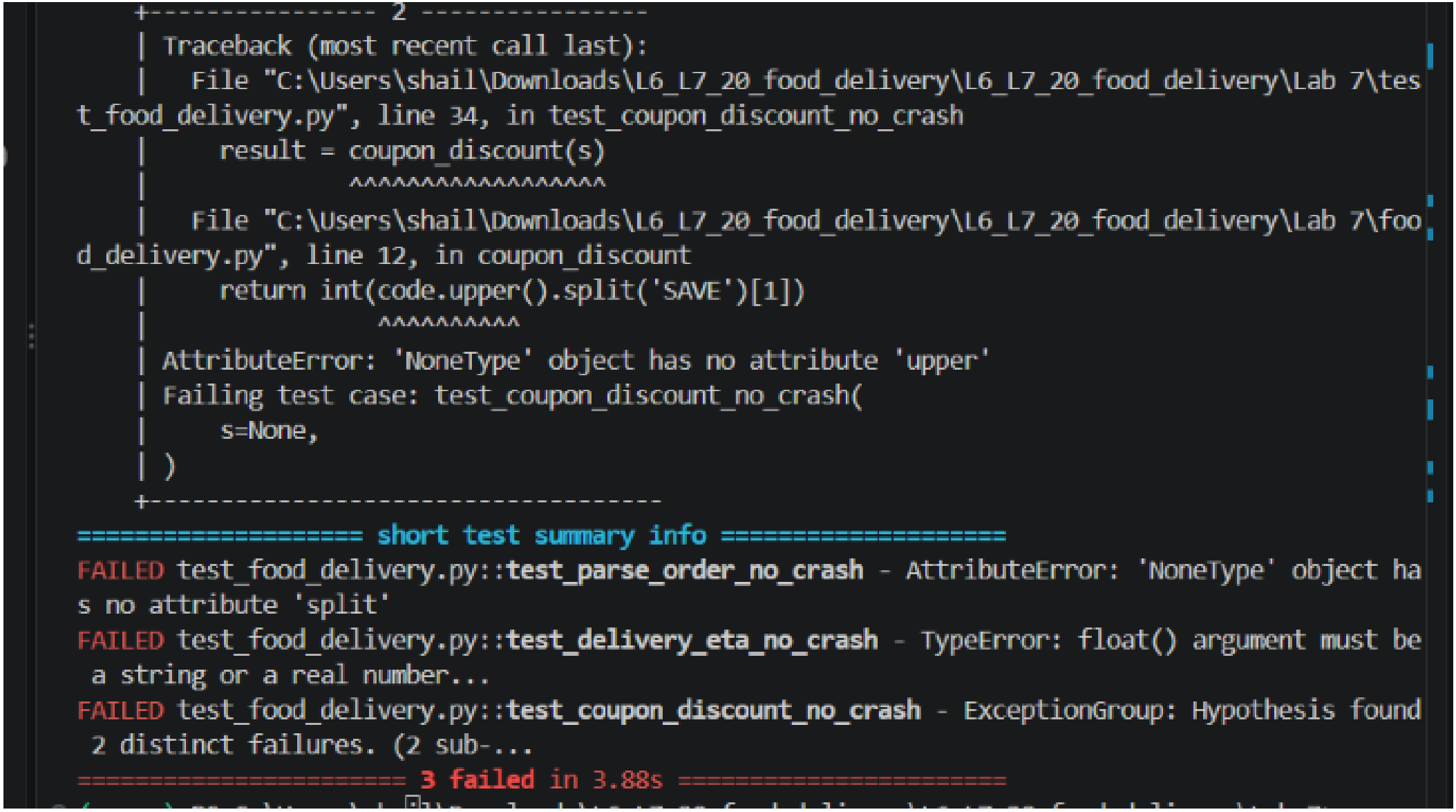
    def delivery_eta(distance_km):
>       return int(15 + float(distance_km) * 4)
                        ^^^^^^^^^^^^^^^^^^
E       TypeError: float() argument must be a string or a real number, not 'NoneType'
E       Failing test case: test_delivery_eta_no_crash(
E           s=None,
E       )
```

5. Fuzz test failure summary



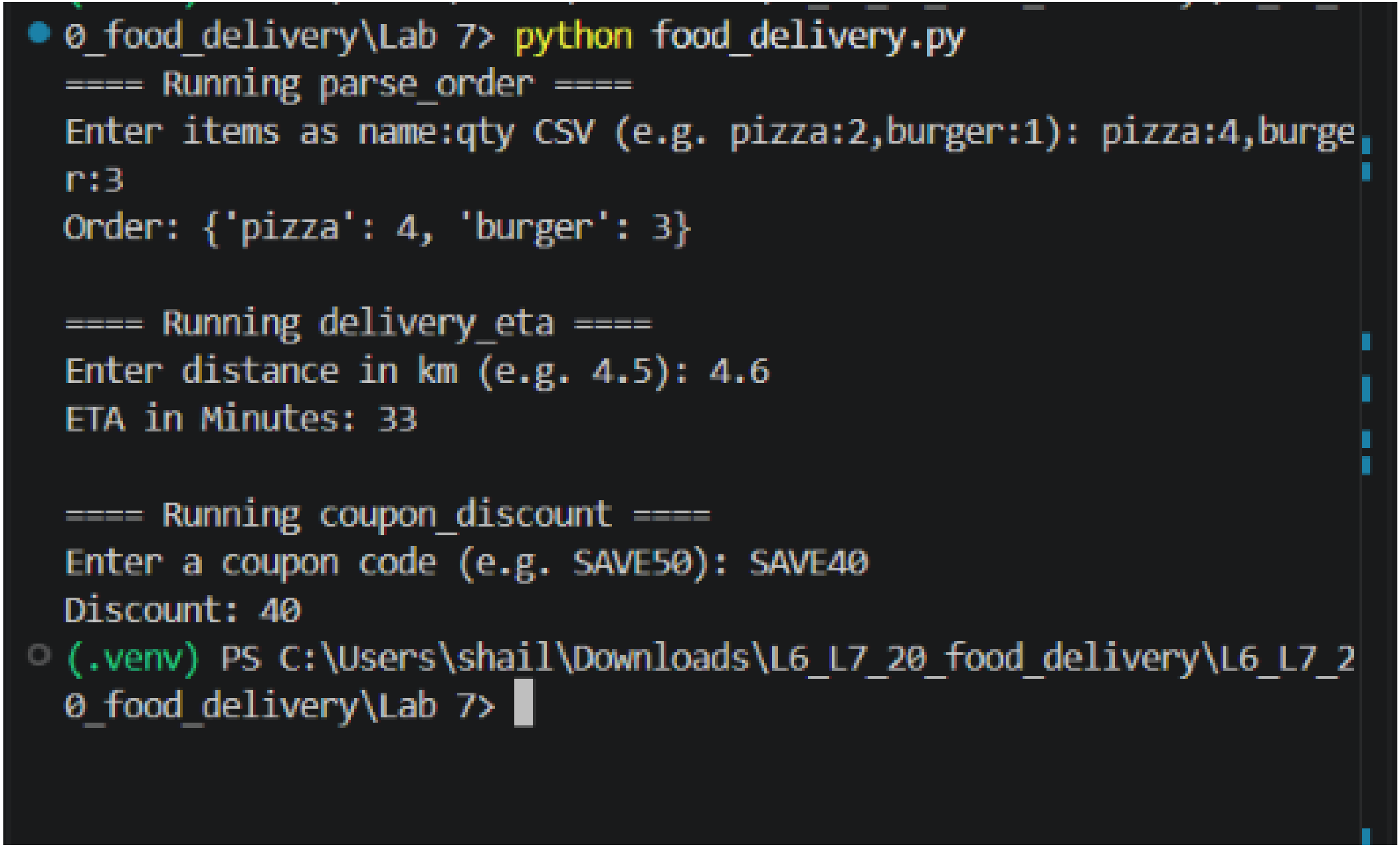
```
food_delivery.py:9: TypeError
_________________ test_coupon_discount_no_crash _________________
  + Exception Group Traceback (most recent call last):
  |   File "C:\Users\shail\Downloads\L6_L7_20_food_delivery\L6_L7_20_food_delivery\Lab 7\test_
food_delivery.py", line 32, in test_coupon_discount_no_crash
  |     def test_coupon_discount_no_crash(s):
  |                    ^^^
  | ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\shail\Downloads\L6_L7_20_food_delivery\L6_L7_20_food_delivery\Lab 7\tes
t_food_delivery.py", line 34, in test_coupon_discount_no_crash
    |     result = coupon_discount(s)
    |              ^^^^^^^^^^^^^^^^^
    |   File "C:\Users\shail\Downloads\L6_L7_20_food_delivery\L6_L7_20_food_delivery\Lab 7\foo
d_delivery.py", line 12, in coupon_discount
    |     return int(code.upper().split('SAVE')[1])
    |                ~~~~~~~~~~~~~~~~~~~~~~~~~~~~~^^^
    | IndexError: list index out of range
    | Failing test case: test_coupon_discount_no_crash(
    |     s='',
    | )
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\shail\Downloads\L6_L7_20_food_delivery\L6_L7_20_food_delivery\Lab 7\tes
t_food_delivery.py", line 34, in test_coupon_discount_no_crash
    |     result = coupon_discount(s)
    |              ^^^^^^^^^^^^^^^^^
    |   File "C:\Users\shail\Downloads\L6_L7_20_food_delivery\L6_L7_20_food_delivery\Lab 7\foo
d_delivery.py", line 12, in coupon_discount
```

6. Additional original failure evidence

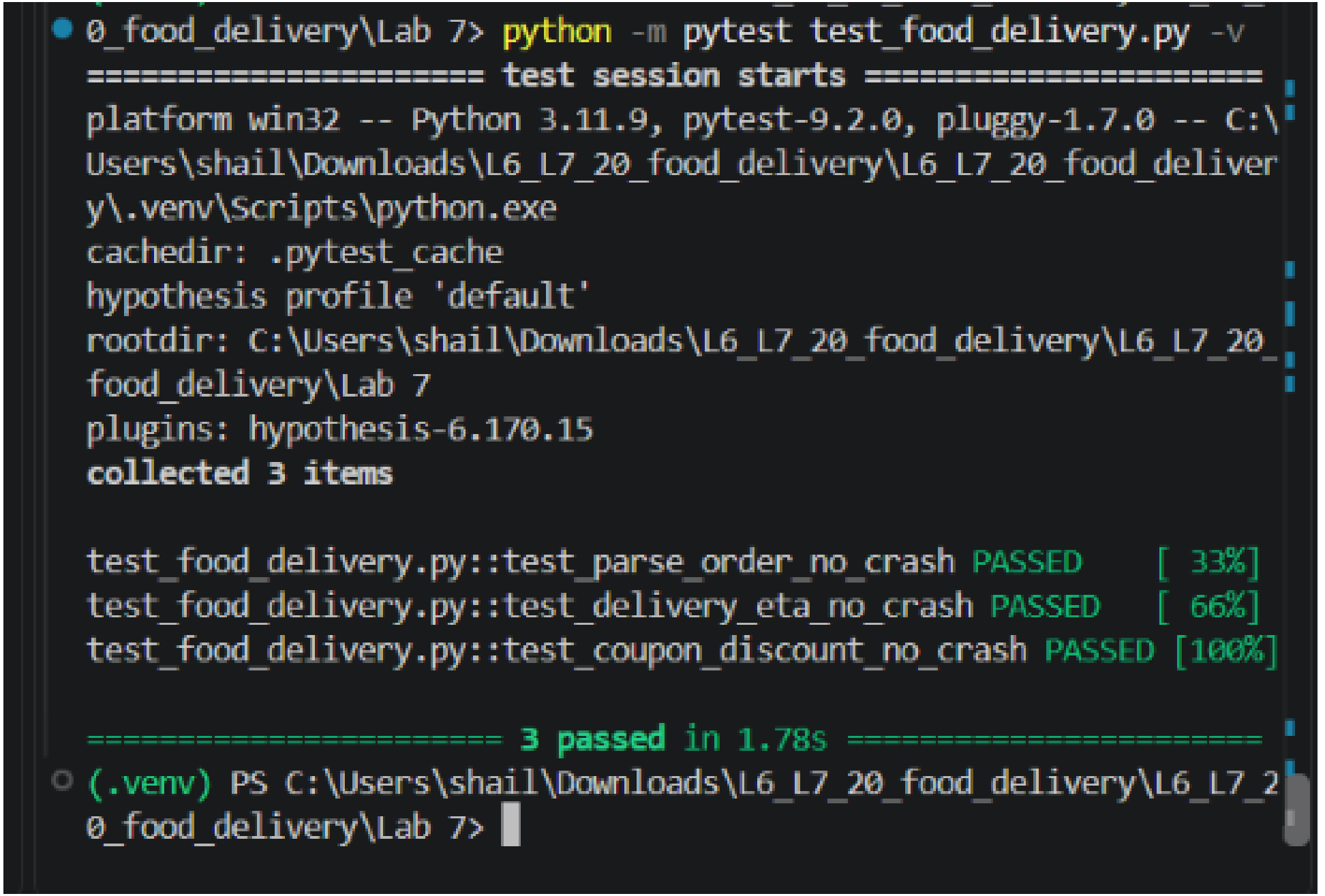


```
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\shail\Downloads\L6_L7_20_food_delivery\L6_L7_20_food_delivery\Lab 7\tes
t_food_delivery.py", line 34, in test_coupon_discount_no_crash
    |     result = coupon_discount(s)
    |              ^^^^^^^^^^^^^^^^^
    |   File "C:\Users\shail\Downloads\L6_L7_20_food_delivery\L6_L7_20_food_delivery\Lab 7\foo
d_delivery.py", line 12, in coupon_discount
    |     return int(code.upper().split('SAVE')[1])
    |                ^^^^^^^^^^
    | AttributeError: 'NoneType' object has no attribute 'upper'
    | Failing test case: test_coupon_discount_no_crash(
    |     s=None,
    | )
    +------------------------------------
================== short test summary info ==================
FAILED test_food_delivery.py::test_parse_order_no_crash - AttributeError: 'NoneType' object ha
s no attribute 'split'
FAILED test_food_delivery.py::test_delivery_eta_no_crash - TypeError: float() argument must be
 a string or a real number...
FAILED test_food_delivery.py::test_coupon_discount_no_crash - ExceptionGroup: Hypothesis found
 2 distinct failures. (2 sub-...
==================== 3 failed in 3.88s =====================
```

7. Fixed Lab 7 application execution



8. Hypothesis tests after fixes – 3 tests passed



9. Manual verification of the fixed functions

